

University of Toronto
Scarborough Campus
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CSC C63 Midterm Test
Instructor: Vassos Hadzilacos

Aids allowed: One 8.5×11 handwritten, non-photocopied ‘cheat sheet’ (may be written on both sides)

Duration: Two hours

READ THE INSTRUCTIONS CAREFULLY

- There should be 12 pages in this exam booklet, including this cover page.
- Answer all questions.
- Put all answers in this booklet, in the spaces provided.
- For rough work, use the backs of the pages; *these will not be graded*.
- The last two pages are blank and can be used for rough work or for overflow. *Do not detach them* from the booklet. If you use them for overflow, you must *clearly indicate this* on the page(s) containing the question(s) whose answer(s) you are providing in the overflow pages.
- In your answers you may use any result discussed in this course or its prerequisites merely by naming or describing it. In particular, you may use any of the facts regarding undecidable or unrecognizable sets listed on page 2.
- Good luck!

LIST OF SOME UNDECIDABLE OR UNRECOGNIZABLE SETS:

- $\text{UNIV} = \{\langle M, x \rangle : \text{TM } M \text{ accepts input } x\}$ is undecidable but recognizable; thus $\overline{\text{UNIV}}$ is unrecognizable.
- $\text{HALT} = \{\langle M, x \rangle : \text{TM } M \text{ halts on input } x\}$ is undecidable but recognizable; thus $\overline{\text{HALT}}$ is unrecognizable.
- $\overline{\text{EMPTY}} = \{\langle M \rangle : \text{TM } M \text{ accepts some input}\}$ is undecidable but recognizable; thus EMPTY is unrecognizable.
- $\text{HET} = \{\langle M \rangle : \text{TM } M \text{ halts on empty tape}\}$ is undecidable but recognizable; thus $\overline{\text{HET}}$ is unrecognizable.
- $\text{FIN} = \{\langle M \rangle : \text{TM } M \text{ recognizes a finite language}\}$ and $\overline{\text{FIN}}$ are both unrecognizable.
- $\text{REG} = \{\langle M \rangle : \text{TM } M \text{ recognizes a regular language}\}$ and $\overline{\text{REG}}$ are both unrecognizable.

QUESTION 1. (10 marks)

Consider a Turing machine M whose set of states includes the states q and p , neither of which is the accept or reject state, whose tape alphabet includes the symbols 0, 1, and the blank $\sqcup$, and whose transition function includes the following transitions:

$$\begin{array}{ll} \delta(q, 0) = (p, 1, L) & \delta(p, 0) = (q, 1, L) \\ \delta(q, 1) = (q, 0, R) & \delta(p, 1) = (q, 1, R) \\ \delta(q, \sqcup) = (p, 1, L) & \delta(p, \sqcup) = (q, 1, L) \end{array}$$

Show the next four configurations in the computation of M that starts with configuration $q01$.

$$q01 \vdash_M \boxed{} \vdash_M \boxed{} \vdash_M \boxed{} \vdash_M \boxed{}$$

QUESTION 2. (10 marks)

We have been observing a Turing machine with 5 states (including the accept and reject states) and 7 tape symbols operate on an input of length 16. If in each of the moves we have observed so far ***the tape head has not moved to the right***, what is the minimum number of moves after which we can conclude that the machine will never halt on that input? Circle one of the choices below; do not justify your answer.

- (a) 1
- (b) 4
- (c) 8
- (d) 11
- (e) 17
- (f) 22
- (g) 36

QUESTION 3. (30 marks)

For each of the following statements indicate whether it is true or false, and *briefly but clearly* justify your answer.

a. The set of functions from $\mathbb{N}$ to $\mathbb{N}$ that are non-increasing is countable. (The function $f: \mathbb{N} \rightarrow \mathbb{N}$ is non-increasing if for every $n \in \mathbb{N}$, $f(n) \geq f(n+1)$.)

Circle the correct answer: True | False

Justification:

b. The set $\{\langle M \rangle: \mathcal{L}(M) \text{ is decidable}\}$ is decidable.

Circle the correct answer: True | False

Justification:

c. For every language A , if there is an enumerator for A , then there is an enumerator for $\overline{A}$.

Circle the correct answer: True | False

Justification:

d. If A is a decidable language then every subset of A is also decidable.

Circle the correct answer: True | False

Justification:

e. For every language $A \in \Sigma^*$, $A \leq_T \overline{A}$. (Recall that $\leq_T$ is the Turing-reduces relation between languages.)

Circle the correct answer: True | False

Justification:

f. The language $\text{SUBSET} = \{\langle M_1, M_2 \rangle : M_1 \text{ and } M_2 \text{ are Turing machines such that } \mathcal{L}(M_1) \subseteq \mathcal{L}(M_2)\}$ is in the set Π_2 of the Arithmetic Hierarchy.

Circle the correct answer: True | False

Justification:

QUESTION 4. (20 marks)

Let A be a recognizable but undecidable language, M_A be a Turing machine that recognizes A , and B be the set of inputs on which M_A loops (i.e., it does **not** halt). Prove that B is unrecognizable.

Proof:

QUESTION 5. (20 marks)

In this question M_1 and M_2 are Turing machines that compute (possibly partial) functions from Σ^* to Σ^* ; and if y is a string, y^R denotes the reverse string of x . Let

$$C = \{ \langle M_1, M_2 \rangle : \text{for some } y \in \Sigma^* \text{ at least one of } M_1 \text{ and } M_2 \text{ outputs } y^R \}.$$

a. Give a *deterministic* Turing machine that recognizes C . You do not need to prove that your Turing machine is correct. For less credit, reason differently to argue that C is recognizable.

Recognizer for C :

b. Prove that C is undecidable. State your reduction clearly and briefly explain why it is correct.

Proof:

[continued on the next page]

[additional space for part (b), if needed]

QUESTION 6. (30 marks)

Consider the following set, where M' is a Turing machine with input alphabet Σ , and $y \in \Sigma^*$:

$$D = \{\langle M' \rangle : \mathcal{L}(M') = \{y : |y| \text{ is odd}\}\}$$

(i.e., D is the set of codes of Turing machines that recognize the set of odd-length strings).

Prove that D and its complement $\overline{D}$ are both unrecognizable. State the reduction(s) you will use; for each reduction you use, describe it clearly, and briefly explain why it is correct.

Answer:

[more space, if needed, on the next page]

[additional space for Question 6, if needed]

THE END

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